

Introduction to Partial Differential Equations

An equation involving one or more partial derivatives of an unknown function of two or more independent variables is called a partial differential equation (PDE).

Example

$x \frac{\partial Z}{\partial x} + \frac{\partial Z}{\partial y} = x$ is a partial differential equation in which Z is the dependent variable and x and y are independent variables.

Order of a partial differential equation

The order of the highest derivative occurring in the partial differential equation.

$\frac{\partial^2 Z}{\partial x^2} - \frac{\partial Z}{\partial y} = x$ is second order.

Degree of a partial differential equation

The degree of the highest derivative which occurs in it after the equation has been rationalized.

Linear and nonlinear partial differential equations

A partial differential equation is said to be linear if the dependent variable and its partial derivatives occur only in the first degree and are not multiplied.

If $Z = Z(x, y)$ is a function of two independent variables x and y , then

$$2 \frac{\partial Z}{\partial x} - 3 \frac{\partial Z}{\partial y} = 0$$

is a homogeneous linear partial differential equation with constant coefficients.

$$\frac{\partial^2 Z}{\partial x^2} - 2x \frac{\partial^2 Z}{\partial x \partial y} + \frac{\partial^2 Z}{\partial y^2} = x + y$$

is a linear partial differential equation with variable coefficients.

$$\frac{\partial^2 Z}{\partial x^2} - 2x \frac{\partial^2 Z}{\partial x \partial y} + \left(\frac{\partial^2 Z}{\partial y^2} \right)^2 = x$$

is a nonlinear partial differential equation with variable coefficients

Formation of a partial differential equation by the elimination of arbitrary constants

Consider the equation

$$F(x, y, Z, a, b) = 0$$

where a and b are arbitrary constants. Eliminating two constants a and b we obtain an equation of the form $f\left(x, y, Z, \frac{\partial Z}{\partial x}, \frac{\partial Z}{\partial y}\right) = 0$ is a required partial differential

equation of first order. Similarly if there are more arbitrary constants elimination will give partial differential equations of higher order.

Example 1

Consider the equation

$$Z = x + by \quad \text{---(1)}$$

here b is an arbitrary constant.

Differentiating (1) partially w.r.t ' x ' we get $\frac{\partial Z}{\partial x} = 1 \quad \dots (2)$

Differentiating (1) partially w.r.t ' y ' we get $\frac{\partial Z}{\partial y} = b \dots (3)$

Eliminating b we get $Z = x + y \frac{\partial Z}{\partial y}$ does not contain arbitrary constant is a required partial differential equation. Since equation (2) does not contain arbitrary constant is also a partial **differential equation**.

Example 2

Consider the equation

$$Z = b(x + y) \quad \dots(1)$$

here b is an arbitrary constant.

Differentiating (1) partially w.r.t ' x ' we get $\frac{\partial Z}{\partial x} = b \dots (2)$

Differentiating (1) partially w.r.t ' y ' we get $\frac{\partial Z}{\partial y} = b \dots (3)$

Eliminating b we get $Z = (x + y) \frac{\partial Z}{\partial x}$ or $Z = (x + y) \frac{\partial Z}{\partial y}$